

Money maths, made readable.

Write down the assumptions first. A correct formula applied to the wrong question can still give a misleading answer.

Rates are fictional

Calculator friendly

Euros for illustration

Percentages: a change has a starting point

$$\text{Percentage change} = (\text{new amount} - \text{old amount}) \div \text{old amount} \times 100\%$$

A price rises from €40 to €46. The change is $€6 \div €40 \times 100\% = 15\%$. A 15% fall from €46 gives €39.10, not €40: the second percentage uses a different starting amount.

Simple interest: interest on the starting amount

$$A = P(1 + rt)$$

P is the initial principal, **r** the annual rate written as a decimal, and **t** years. With €1,000 at 4% simple interest for 3 years: $A = 1,000(1 + 0.04 \times 3) = \text{€1,120}$.

Compound interest: interest also earns interest

$$A = P(1 + r)^t \quad [\text{annual compounding}]$$

With the same €1,000 and 4% rate compounded annually for 3 years: $A = 1,000 \times 1.04^3 = \text{€1,124.86}$. The first year earns €40; the next earns interest on €1,040.

END OF YEAR	SIMPLE INTEREST	ANNUAL COMPOUNDING
1	€1,040.00	€1,040.00
2	€1,080.00	€1,081.60
3	€1,120.00	€1,124.86

Mind the rate convention. If **r** is a nominal annual rate compounded **m** times a year, use $A = P(1 + r/m)^{mt}$. The effective annual rate is $(1 + r/m)^m - 1$. A 12% nominal annual rate compounded monthly gives about 12.68% effective annual growth, not 12%.

Assumptions: fixed rates, no deposits or withdrawals, no fees or taxes. These examples show arithmetic, not guaranteed investment returns. For irregular periods or changing rates, match the calculation to the actual contract or cash flows.

Saving and borrowing use the same clock.

Adding the same amount each month

Let **P** be the initial balance, **c** the payment into savings at each month's end, **i** the monthly interest rate, and **n** the number of months.

$$A = P(1 + i)^n + c[(1 + i)^n - 1] / i \quad (i \neq 0)$$

$$A = P + cn \quad (i = 0)$$

Example: start with €500, add €50 at each month's end, and use a 4% nominal annual rate compounded monthly for 3 years. Then $i = 0.04/12$ and $n = 36$. The final balance is **€2,472.71**. Contributions are €500 + 36 × €50 = €2,300; interest is **€172.71**.

If deposits happen at each month's beginning, multiply only the contribution term by $(1 + i)$. If the quoted rate is effective annual, the equivalent monthly rate is $(1 + r)^{1/12} - 1$, not $r/12$.

Repaying a fixed-rate loan

Let **L** be the loan amount, **i** the monthly rate and **n** the number of equal month-end payments. The payment **M** that repays the loan is:

$$M = Li / [1 - (1 + i)^{-n}] \quad (i > 0)$$

$$M = L/n \quad (i = 0)$$

Example: borrow €1,200 at 12% nominal annual interest, compounded monthly, and repay over 12 months. $i = 0.01$. The unrounded monthly payment is €106.618546...; total repayment is approximately **€1,279.42**, including **€79.42** interest. With cents rounding, the final payment may need a small adjustment.

Check the first month by hand

STEP	AMOUNT
Starting balance	€1,200.00
Interest: €1,200 × 1%	€12.00
Subtract first payment, rounded	€106.62
Remaining balance	€1,105.38

Why some debts grow. The monthly update is $B_{\text{next}} = B(1 + i) - M$. At a balance of €1,200 and $i = 1\%$, a €10 payment is below €12 interest. The balance becomes €1,202 before any fees. Paying something does not always mean owing less.

Loan model: fixed rate, equal monthly periods, no fees, no new borrowing and month-end payments. APR or another total-cost measure can include fees and follow a different convention; do not substitute it blindly as the nominal rate. Read the terms and compare total repayment as well as the monthly amount.

What money buys. What risk means.

Inflation changes the price level

For a constant annual inflation rate π , a basket costing C_0 today has modelled future cost C_t . A fixed future sum F has purchasing power expressed in today's euros:

$$C_t = C_0(1 + \pi)^t$$

$$\text{Today's-euro value of } F = F / (1 + \pi)^t$$

At 3% a year for 5 years, a €100 basket becomes **€115.93**. A fixed €100 at that time buys what **€86.26** buys today. These are two different questions; keep them separate.

A return after inflation

$$\text{Real return} = (1 + \text{nominal return}) / (1 + \text{inflation rate}) - 1$$

If a balance grows 4% while prices rise 3% over the same year, the real return is $1.04/1.03 - 1 \approx \mathbf{0.97\%}$. Subtracting 3% from 4% gives a useful approximation here, not the exact result. Fees and taxes can reduce what remains.

Expected value is an average, not a promise

$$E[X] = p_1x_1 + p_2x_2 + \dots + p_kx_k$$

For mutually exclusive outcomes covering all possibilities, probabilities must sum to 1. A fictional game costs €2 and pays a €10 prize with probability 0.1; otherwise it pays nothing. Net outcomes are +€8 (probability 0.1) and –€2 (probability 0.9). Expected net gain is $0.1 \times 8 + 0.9 \times (-2) = \mathbf{-€1 \text{ per play}}$.

You cannot receive exactly –€1 in one play. It is the probability-weighted average. The example illustrates expected value; it is not an invitation to gamble.

Diversification: count the risks, not just the names

Splitting a hypothetical €100 equally across two assets with returns +10% and –6% produces €55 + €47 = €102, a **2% portfolio return**. In general, the one-period portfolio return is $\sum w_j r_j$, with weights summing to 1.

This single outcome does not prove lower risk. Risk reduction depends on how returns move together. Ten closely related holdings can still fall together. Diversification can reduce concentration risk; it cannot guarantee a profit or eliminate market risk.

Before trusting any result: check the units, time period, payment timing, rate convention, fees, rounding and whether the assumptions fit the question.

Background: [ECB: What is inflation?](#); [Investor.gov: Compound Interest Calculator](#). Formula derivations and numerical examples are included here for teaching; rates are not forecasts or product offers.